Reducing Implicit Misalignment in Extreme Universal Domain Adaptation

Hung-Chieh Fang
National Taiwan University
b09902106@csie.ntu.edu.tw

Po-Yi Lu
National Taiwan University
d09944015@csie.ntu.edu.tw

Hsuan-Tien Lin
National Taiwan University
htlin@csie.ntu.edu.tw

Abstract

Universal Domain Adaptation (UniDA) seeks to address the challenges of transferring knowledge from a labeled source domain to an unlabeled target domain without knowing how much their label sets overlap with each other. The goal of UniDA is to achieve robust performance across varying class ratios. However, existing UniDA approaches struggle in scenarios with many non-overlapping source classes, a situation we refer to as *Extreme UniDA*. The unsatisfactory performance can be attributed to misaligning target instances to source instances in the learned representation space. In this work, we study the misalignment issue deeper by dissecting it into two parts: *implicit misalignment* and *explicit misalignment*. Typically, existing UniDA approaches start from an initial representation space fine-tuned with the source data and then update the representation space by pulling the two domains together in the alignment process. Implicit misalignment happens at the initial representation space, and explicit misalignment emerges in the alignment process. We identify that existing approaches are ineffective for Extreme UniDA because they mainly focus on resolving explicit misalignment, while implicit misalignment in Extreme UniDA is the dominating cause due to the extensive non-overlapping source classes. To address this, we propose a source-label-free target incorporation module that leverages self-supervised learning to **integrate target data independently of source data before the alignment process**. Our experimental results and analysis demonstrate the effectiveness of our proposed approach in Extreme UniDA.

1 Introduction

Advancements in deep neural networks have heavily relied on the assumption that abundant data follows the same distribution, making it difficult for these networks to generalize well on unseen data sampled from different distributions. Unsupervised Domain Adaptation (UDA) [21] addresses this issue by transferring knowledge from a source domain with a known distribution to a target domain with a different distribution. Nevertheless, most UDA methods [10, 18, 13] operate under the strong assumption that the label sets of the source domain ($\mathcal{C}_s$) and the target domain ($\mathcal{C}_t$) are the same ($\mathcal{C}_s = \mathcal{C}_t$), limiting the applicability in real-world scenarios.

To overcome this limit, more flexible setups such as Open-set Domain Adaptation ($\mathcal{C}_s \subset \mathcal{C}_t$) and Partial Domain Adaptation ($\mathcal{C}_s \supset \mathcal{C}_t$) have been studied [22, 3], addressing scenarios where the target label set has more or fewer classes than the source label set. Universal Domain Adaptation (UniDA) further loosens the setup by not assuming $\mathcal{C}_s$ and $\mathcal{C}_t$ to have any containment relation. That is, while

Preprint. Under review.

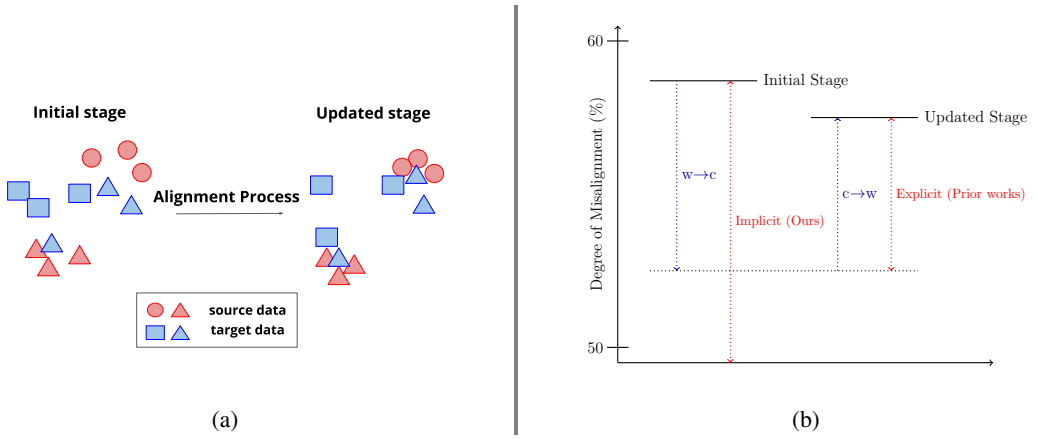


Figure 1: **Illustration of misalignment in the training process.** (a) The evolution of feature space before and after the alignment process. Different shape indicates different class. (b) The change of misalignment. w→c indicates wrong instances aligned to correct instances, while c→w indicates the inverse. We dissect the misalignment into *implicit misalignment* (Implicit) and *explicit misalignment* (Explicit).

$\mathcal{C}_s$ and $\mathcal{C}_t$ overlaps on some unknown common classes where the domain adaption takes place, each of them can have its own private classes. The objective of UniDA is to classify the target examples either as the target-private super-class, or as one of the (target-)common classes.

The ultimate goal of UniDA is to achieve consistent performance regardless of the ratio of private classes. However, previous experimental settings often overlook scenarios where the proportion of source-private classes is substantial, a situation we refer to as *Extreme UniDA*. Extreme UniDA is common in real-world applications, where a classifier may be trained on a large dataset with numerous classes, such as ImageNet [8], but used for inference on a much smaller, specific application dataset in the target domain. We find that existing approaches often fall short in Extreme UniDA due to severe misalignment between source and target instances in the representation space. The observation motivates us to investigate how misalignment is exacerbated in Extreme UniDA.

In UniDA, the initial stage typically involves fine-tuning a feature extractor using source data. In the subsequent stage, a distribution-matching loss, such as domain-adversarial training [10, 18, 13], or self-training methods [17, 20], is employed to align the source and target domains, as illustrated in Fig. 1a. In Fig. 1b, we illustrate how misalignment can occur during the training process. At the initial stage, the representation space shows misalignment between source and target data due to distribution differences, which we refer to as *implicit misalignment*. After the alignment process, the feature space in the updated stage may correct some instances (w→c), while other instances may become incorrectly aligned (c→w), which we term *explicit misalignment*. Existing UniDA methods [31, 9, 16, 28, 6] often address the UniDA problem with partial alignment, where private instances is assigned with lower weights in the alignment process. These approaches mainly focus on mitigating explicit misalignment, as they prevent private instances from aligning with common instances. However, we found that Extreme UniDA exhibits severe implicit misalignment, which cannot be easily corrected by mitigating explicit misalignment.

In this work, we propose to **incorporate target information prior to the alignment process** at the initial stage. To implement this idea, we develop a source-label-free target incorporation module based on self-supervised learning (SSL). This module aims to separate instances from different domains while aggregating instances with the same label, thereby reducing implicit misalignment.

Furthermore, our extensive study reveals an interesting side finding that SSL can also be coupled with the alignment process to alleviate explicit misalignment. The results and ablation studies support that SSL would effectively integrate the target information in handling implicit misalignment for extreme UniDA problems. The main contributions can be summarized as follows:

- We are the first to study *Extreme UniDA* problems that reveal the limitation of existing UniDA methods and expand to a new border of the UniDA field.

- We dissect the misalignment issues into *implicit misalignment* and *explicit misalignment*, then point out that *implicit misalignment* is an overlooked issue that accounts for the unsatisfactory performance of existing UniDA methods.
- We adopt self-supervised learning in our proposed source-label-free target incorporation module to effectively reduce implicit misalignment at the initial stage, which outperforms the state-of-the-art in Extreme UniDA.

2 Dissection of Misalignment in Universal Domain Adaptation

Universal Domain Adaptation (UniDA) [31] has a labeled source domain $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ and an unlabeled target domain $\mathcal{D}_t = \{x_i^t\}_{i=1}^{n_t}$ accessible at training time. The label sets are denoted as $\mathcal{C}_s$ for the source domain and $\mathcal{C}_t$ for the target domain, respectively. $\mathcal{C} = \mathcal{C}_s \cap \mathcal{C}_t$ represents the common label set shared by both domains. Let $\overline{\mathcal{C}}_s = \mathcal{C}_s \setminus \mathcal{C}$ and $\overline{\mathcal{C}}_t = \mathcal{C}_t \setminus \mathcal{C}$ represent the source-private label set and the target-private label set, that a label only occurs at one of two domains. The goal of UniDA is to classify target examples into $|\mathcal{C}| + 1$ classes, where target-private examples are regarded as one unknown class.

2.1 A Measurement of Misalignment in UniDA

The primary objective of UniDA is to perform effectively under arbitrary class ratios. Despite the emerging research in UniDA with diverse experimental settings, prior studies often overlook extreme scenarios where the proportion of source-private classes is significant, resulting in poor performance. The performance degradation can be attributed to misalignment between source and target data in the representation space, which hinders the source classifier from forming an accurate decision boundary, resulting in the incorrect classification of target instances.

To understand how misalignment occurs during the training process, it is crucial to develop a measurement method. We introduce a new metric, the Cross-Domain Alignment Score (CDAS), to quantify the degree of misalignment between domains. For each instance (x_i^t, y_i^t) in the target domain, we identify the set $N_{x_i^t}^s$ of k-nearest neighbors in the source domain based on a pre-defined distance metric. The score is calculated as:

$$\text{CDAS} = \frac{1}{|\mathcal{D}_t|} \sum_{(x_i^t, y_i^t) \in \mathcal{D}_t} [\![\text{mode}\{y_j^s \mid (x_j^s, y_j^s) \in N_{x_i^t}^s\} = y_i^t]\!], \tag{1}$$

where mode returns the most frequent number in a set. The rationale behind our score design is discussed in Section 4.3. The following section further explores how misalignment forms and becomes exacerbated in extreme cases.

2.2 Implicit and Explicit Misalignment

Implicit Misalignment. At the initial stage of the training process, we minimize the classification loss in the source domain $\mathcal{L}_s$, which is defined as follows:

$$\mathcal{L}_s(f_\theta, c_\theta) = \mathbb{E}_{(x,y) \in \mathcal{D}_s} \mathbb{CE}(y, c_\theta(f_\theta(x))), \tag{2}$$

where the parameterized model consists of a feature extractor f_θ and a label classifier c_θ, $\mathbb{CE}$ is the cross-entropy loss function. Due to the differences in source and target distributions, the feature extractor f_θ tends to be dominated toward the source data, causing misalignment between the source and target data. We refer to this type of misalignment as the *implicit misalignment*.

Explicit Misalignment. In the alignment process, prior UDA works leverage domain-adversarial training [10, 18, 13] or self-training [17, 20] to address the gap between the source and target domains. A classical approach [10] involves aligning domains by confusing a domain classifier d_θ:

$$\mathcal{L}_{\text{adv}}(f_\theta, d_\theta) = \mathbb{E}_{x \in \mathcal{D}_s} \mathbb{CE}(0, d_\theta(f_\theta(x))) + \mathbb{E}_{x \in \mathcal{D}_t} \mathbb{CE}(1, d_\theta(f_\theta(x))). \tag{3}$$

While these methods may align features of instances across two domains, there is a risk that some instances are wrongly aligned. We refer to this issue as *explicit misalignment*, where instances from the source domain are incorrectly aligned with those from the target domain, degrading the overall alignment quality.

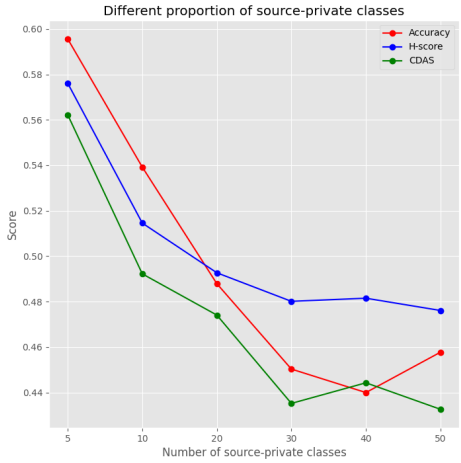


Figure 2: Performance and Cross-Domain Alignment Score (CDAS) with diverse proportion of source-private classes. H-score is the harmonic mean of accuracy in common examples and accuracy in private samples, which is widely used in UniDA literature.

How does Extreme UniDA exacerbate implicit and explicit misalignment? In Fig. 2, we assess the performance of CDAS using a source classifier at the initial stage across various proportions of source-private classes. We can observe that performance and CDAS decrease with an increase in source-private classes, indicating a worsening implicit misalignment. Additionally, the non-overlapping label space between domains can inadvertently cause common-class features to merge with private-class features during the alignment process, resulting in more pronounced explicit misalignment.

2.3 Limitation of Prior Works

Prior works [31, 9, 16, 28, 6] often address UniDA problems with partial alignment. PADA [3] initiated the concept of partial alignment, which carefully designed a weighting function for the source domain $w_s(x)$ and the target domain $w_t(x)$ to align the common label set $\mathcal{C}$. The weighting function assigns higher weights to examples likely to belong to the common label set $\mathcal{C}$ in the adversarial loss. We can reformulate the loss function in (3):

$$\mathcal{L}^{w}_{\text{adv}}(f_\theta, d_\theta) = \mathbb{E}_{x\in\mathcal{D}_s} w_s(x) \cdot \mathbb{CE}(0, d_\theta(f_\theta(x))) + \mathbb{E}_{x\in\mathcal{D}_t} w_t(x) \cdot \mathbb{CE}(1, d_\theta(f_\theta(x))). \tag{4}$$

This technique aims to prevent features of common-class instances from aligning with features of private-class instances during the alignment process, focusing on reducing the explicit misalignment problem. However, we found that existing approaches still struggle in Extreme UniDA, where implicit misalignment is substantial.

3 Methodology

In this section, we categorize the training process into two parts with two stages mentioned in Figure 1. We first present the *source-label-free target incorporation* module for the implicit misalignment that occurs in the initial stage in Section 3.1. Then, we introduce the *enhanced alignment process* module for the explicit misalignment in the updated stage in Section 3.2.

3.1 Source-Label-Free Target Incorporation for Implicit Misalignment

The implicit misalignment causes the feature of target instances to be entangled with incorrect source instances at the initial stage, which hinders the following alignment process during training. If we aim to extricate the entangled features from incorrect instances, we should rely on more information beyond the source data. A central question we explore is *how to integrate target data*

before the alignment process effectively. Self-training (ST) is a common technique for incorporating target information, but it still depends on the source classifier, which directly utilizes source-label information. This dependency can lead to incorrect pseudolabels for target data in cases of severe misalignment.

In this work, we adopt self-supervised learning (SSL) for target incorporation, which extracts information across the source and target domains without any label information, to escape the dilemma of existing approaches for the extreme UniDA problems. We can formulate the self-supervised loss function before the alignment process as

$$\mathcal{L}_{\text{ssl}}(f_\theta) = \mathbb{E}_{x \in \mathcal{D}_s \cup \mathcal{D}_t} \| f_\theta(\mathcal{T}(x)) - f_\theta(\mathcal{T}'(x)) \|_2^2, \tag{5}$$

where f_θ is the feature extractor, $\mathcal{T}$ and $\mathcal{T}'$ are the random augmentation functions, and $\mathcal{D}_s$ and $\mathcal{D}_t$ are the source data and the target data, respectively. We use SimSiam [7] as the SSL method in our implementation because it does not require negative samples and a large batch size. We further study the effectiveness of SSL in handling implicit misalignment in Section 4.3.2.

3.2 Enhanced Alignment Process for Explicit Misalignment

After obtaining a representation space with less implicit misalignment using the source-label-free target incorporation, we now focus on the alignment process. The alignment process could build on the existing partial alignment methods, which adopt adversarial learning (4) to align source and target domains. However, we know that existing partial alignment methods still suffer from explicit misalignment, which is the main challenge during the alignment process.

In this work, we continually use the SSL for the alignment process to reduce explicit misalignment. Intuitively, SSL can act as a regularizer to alleviate the wrong direction of adversarial loss in the alignment process. We present the benefits of the SSL in addressing explicit misalignment in Section 4.3.2. At last, the objective function of the alignment process can be formulated as

$$\min_{f_\theta, c_\theta, d_\theta} \left[\mathcal{L}_s + \lambda_{\text{ssl}} \cdot \mathcal{L}_{\text{ssl}} - \lambda_{\text{adv}} \cdot \mathcal{L}_{\text{adv}}^w \right], \tag{6}$$

where λ_{ssl} and λ_{adv} are the weighting hyperparameters.

4 Experiments

We present experimental settings, including datasets, existing approaches, and an evaluation metric in Section 4.1. We demonstrate experimental results to compare our method with existing approaches tailored to UniDA in Section 4.2. At last, beyond avoiding the plight of unaware using source-label information, we study the measurement of misalignment and SSL for effectively attacking Extreme UniDA in Section 4.3.

4.1 Experimental Settings

Dataset. We report results on four widely used benchmarks. Office31 [26] contains 31 classes and three domains: Amazon (A), DSLR (D), and Webcam (W), with a total of about 4k images. Office-Home [29] has 65 classes and four domains: Art (A), Product (Pr), Clipart (Cl), and Realworld (Rw), with approximately 15k images. VisDA [23] is a larger dataset with 12 classes from two domains: Synthetic and Real images, totaling around 280k images. DomainNet [24], the largest DA dataset, has 345 classes and six domains, with about 0.6 million images. Following prior works [9, 4, 15], we use only three domains: Real (R), Sketch (S), and Painting (P).

Extreme UniDA setting. Previous studies have explored ratios of private and common classes across different datasets. However, these settings have not adequately revealed scenarios that source-private classes predominate across all classes. Therefore, we experiment on the class ratio with over 70% source-private classes for each dataset (details provided in Appendix A.2), demonstrating the serious implicit misalignment brought by large number of source-private classes.

Evaluation metric. Recall the goal of UniDA is to classify examples to $|C| + 1$ classes, i.e., $|C|$ common classes and one for target-private classes. To ensure our classifier performance well on both common and target-private classes, we adopt H-score [9] as our metric, which calculate the harmonic mean of accuracy on common classes and accuracy on target-private (unknown) classes.

	Office-Home													VisDA
	Ar2Cl	Ar2Pr	Ar2Rw	Cl2Ar	Cl2Pr	Cl2Rw	Pr2Ar	Pr2Cl	Pr2Rw	Rw2Ar	Rw2Cl	Rw2Pr	Avg	S2R
UAN[31]	29.9	36.4	14.1	22.4	20.6	16.4	26.4	25.1	27.3	31.3	24.4	35.4	25.8	41.5
CMU[9]	38.5	43.5	45.7	41.4	41.2	47.5	46.0	46.6	40.3	41.5	38.5	27.2	41.5	34.1
DANCE [28]	34.0	55.5	**82.6**	43.4	44.2	60.1	34.4	20.8	61.2	**65.7**	33.6	61.7	44.6	69.1
UniOT [4]	27.2	32.3	26.6	28.4	29.9	23.2	31.4	29.3	23.0	35.9	34.3	35.3	29.7	49.9
Ours	**47.1**	**74.4**	76.8	**46.4**	**54.3**	**63.5**	**55.8**	**48.5**	**72.3**	57.7	**46.3**	**68.5**	**59.3**	**89.5**

Table 1: H-score (%, ↑) on **Office-Home** and **VisDA**.

	Office							DomainNet						
	A2D	A2W	D2A	D2W	W2A	W2D	Avg	P2R	R2P	P2S	S2P	R2S	S2R	Avg
UAN[31]	24.5	61.8	48.9	64.2	27.9	61.3	48.1	11.9	15.1	14.4	17.2	18.1	11.3	14.6
CMU[9]	76.8	63.8	56.1	77.2	66.3	78.2	69.7	30.1	**42.4**	34.1	24.3	32.2	34.1	32.8
DANCE[28]	49.7	47.9	48.4	54.9	48.9	55.6	50.9	39.4	3.30	11.8	0.90	7.60	35.3	16.4
UniOT[4]	78.8	67.7	**86.1**	66.9	**83.8**	81.0	77.4	38.1	29.8	30.8	29.3	29.1	38.3	32.6
Ours	**87.4**	**74.9**	72.4	**81.3**	74.9	**87.7**	**79.8**	**50.1**	39.2	**35.9**	**32.7**	**34.0**	**49.8**	**40.3**

Table 2: H-score (%, ↑) on **Office** and **DomainNet**.

Existing approaches We evaluate several existing approaches, which open-source code is available, for the extreme UniDA problems. These including Universal Adaptation Network (UAN) [31], Calibrated Multiple Uncertaintie (CMU) [9], Domain Adaptative Neighborhood Clustering via Entropy optimizatio (DANCE) [28] and Optimal Transport for UniDA (UniOT) [4]. These approaches are designed to handle the general UniDA problems, and we aim to compare their performance with ours in the extreme UniDA setting.

Implementation details. We use ResNet-50 [12] as the backbone model for all experiments, which is pre-trained on ImageNet [8]. The optimizer, scheduler, learning rate, batch size, and training steps are consistent with the baselines [31]. Both λ_{ssl} and λ_{adv} are set to 0.5. For the self-supervised loss, we use the same augmentations as SimSiam [7], and a 2-layer linear layer is added as the predictor in the SimSiam component.

4.2 Results of Extreme UniDA

Tables 1 and 2 summarize the results in the extreme UniDA setting across four different DA benchmarks. Our method demonstrates significant improvements over previous approaches in these extreme UniDA settings. Specifically, it outperforms the previous state-of-the-art method, UniOT, by 29.6% on Office-Home, 39.6% on VisDA, and 7.7% on DomainNet. An observation is that our method does not show a clear advantage in the Office dataset. This can be attributed to the simplicity of the small dataset, where the initial feature space has minimal implicit misalignment.

Additionally, we conducted experiments using standard settings to show that our method remains effective in general cases (see Appendix C.1). Our method achieves performance comparable to previous works using the domain-adversarial training paradigm. However, in the Office-Home dataset, our method slightly falls short compared to UniOT, which might be attributed to the advantage of optimal transport in handling non-severe misalignment.

It is important to note that existing approaches rely on source-label information and fail to address implicit misalignment in extreme UniDA scenarios. In contrast, our methods effectively address the issue of misalignment in extreme UniDA scenarios without negatively impacting general cases. We attribute this success to the incorporation of source-label-free information through SSL. In the following section, we will discuss the effectiveness of SSL in handling implicit misalignment and how it enhances the alignment process by mitigating explicit misalignment.

4.3 Discussion

In this section, we discuss our proposed metric and solution for addressing misalignment issues. We first compare the different measurements of misalignment in Section 4.3.1. Then we analyze the effectiveness of SSL to integrate source-label-free target information in handling misalignment in Section 4.3.2.

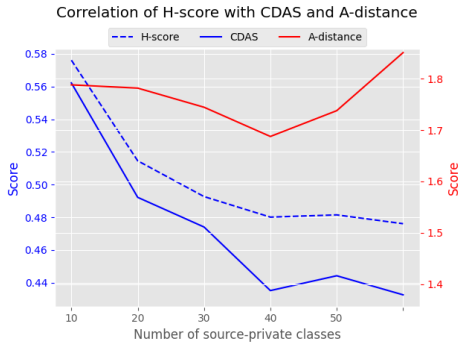


Figure 3: Correlation of H-score with Cross-Domain Alignment Score (CDAS) and $\mathcal{A}$-distance.

Method	H-score
$\mathcal{L}_s$	63.2
$\mathcal{L}_s + \mathcal{L}_{\text{adv}}$	60.0
$\mathcal{L}_s + \mathcal{L}_{\text{ssl}}$	67.4
$\mathcal{L}_s + \mathcal{L}_{\text{ssl}} + \mathcal{L}_{\text{adv}}$	**72.3**

Table 3: Ablation study of different losses.

4.3.1 Measurement of Misalignment

Why choose CDAS over $\mathcal{A}$-distance? To measure the degree of misalignment during the training process, we need a metric that accurately reflects both the results and the misalignment. $\mathcal{A}$-distance [2] is widely used to calculate the discrepancy between two domains (see Appendix A.3 for detailed calculation). However, we observe that $\mathcal{A}$-distance does not strongly correlate with final performance in UniDA, as illustrated in Fig. 3. In contrast, our newly designed CDAS metric provides a more accurate reflection of the results.

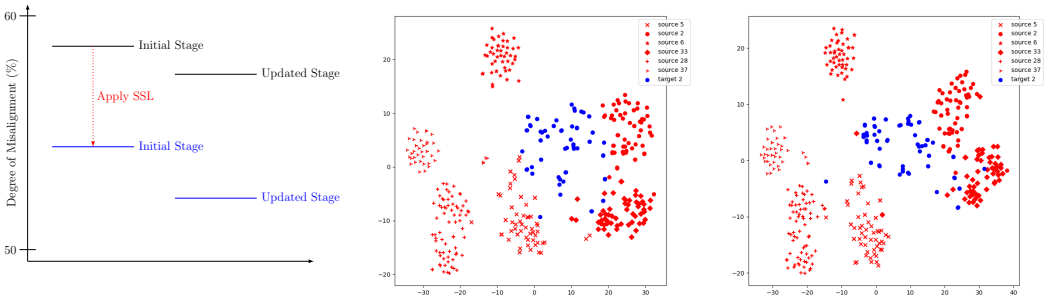
4.3.2 Study on SSL for Extreme UniDA

Ablation study of loss functions. In Table 3, we examine various combinations of our loss functions. We set the source classification loss ($\mathcal{L}_s$) alone without any alignment as the baseline, and demonstrate that directly performing partial alignment ($\mathcal{L}_s + \mathcal{L}_{\text{adv}}$) on feature space with severe implicit misalignment is suboptimal. By contrast, combining source classification loss with self-supervised loss ($\mathcal{L}_s + \mathcal{L}_{\text{ssl}}$) helps mitigate the implicit misalignment and achieve improvements. Additionally, we show that combining SSL with partial alignment ($\mathcal{L}_s + \mathcal{L}_{\text{ssl}} + \mathcal{L}_{\text{adv}}$) can further enhance performance. Therefore, we argue that SSL is crucial in addressing implicit misalignment and can strengthen the alignment process. Next, we will verify the benefits of SSL in handling implicit misalignment at the initial stage and partial alignment in the alignment process.

How SSL mitigate implicit misalignment? As shown in Fig. 4a, applying SSL at the initial stage significantly reduces misalignment, providing a much better initialization. This indicates SSL's effectiveness in addressing implicit misalignment. We further analyze the impact of SSL on the representation space. Fig.4b depicts the feature space of a source classifier at the initial stage, while Fig.4c illustrates the feature space of a source classifier coupled with SSL. It is evident that, initially, the target features from the 2nd class are very close to the source features from the 5th class when using source classification loss alone. After incorporating SSL, the target features from the 2nd class become more aggregated and separate from the source features of the 5th class.

Table 4 quantifies how SSL affects the representation space. The first and fourth rows demonstrate that both $\mathbf{D}^{t\to s}_{\text{correct}}$ and $\mathbf{D}^{t\to s}_{\text{wrong}}$ increase when SSL is integrated, indicating that SSL effectively **separates features from different domains**, regardless of their true class. Additionally, the aggregation score increases with SSL, showing that SSL **enhances the aggregation of target features**. This suggests that SSL can effectively separate features and reduce implicit misalignment, providing a better initialization for the subsequent alignment process.

In Table 4, we further present the results of an ablation study on SSL across different domain combinations. The study shows that applying SSL to either the source or target domain individually helps to separate features from different domains, thereby reducing implicit misalignment and improving performance. Additionally, applying SSL to the target data further enhances feature aggregation within the target domain, supporting our design choice to train SSL on both the source and target data.



(a) (b) (c)

Figure 4: **The effect of applying SSL at initial stage.** (a) Change of degree of misalignment evaluated with CDAS. Black lines are trained with $\mathcal{L}_s$, while blue lines are trained with $\mathcal{L}_s + \mathcal{L}_{\text{ssl}}$. (b) The t-SNE visualization at initial stage trained with $\mathcal{L}_s$. (c) The t-SNE visualization at initial stage trained with $\mathcal{L}_s + \mathcal{L}_{\text{ssl}}$.

Method	**H-score**	$\mathbf{D}^{t\to s}_{\text{correct}}$	$\mathbf{D}^{t\to s}_{\text{wrong}}$	**Agg. in** t (↑)
$\mathcal{L}_s$	44.1	0.9614	0.9491	0.1199
$\mathcal{L}_s + \mathcal{L}^s_{\text{ssl}}$	47.2	0.9697	0.9571	0.1046
$\mathcal{L}_s + \mathcal{L}^t_{\text{ssl}}$	50.2	0.9722	0.9542	0.1487
$\mathcal{L}_s + \mathcal{L}_{\text{ssl}}$	50.7	0.9899	0.9699	0.1537

Table 4: **Ablation study of SSL.** $\mathcal{L}^{\bullet}_{\text{ssl}}$ means applying SSL on source or target data only. $\mathbf{D}^{t\to s}_{\text{correct}}$ measures the pairwise euclidean distances of target instances from the i-th class to source instances from the same class, while $\mathbf{D}^{t\to s}_{\text{wrong}}$ measures the pairwise euclidean distances of target instances from the i-th class to source instances from the closest different classes. **Agg. in** t measures the degree of aggregation in target features of the i-th class (detailed calculations are provided in Appendix A.4).

How SSL mitigate explicit misalignment? To verify if SSL can enhance the alignment process, we conducted an ablation study by removing SSL during the alignment stage. As shown in Table 5, the inclusion of SSL can further reduce explicit misalignment and improve performance. We hypothesize that SSL acts as a regularization on the adversarial loss, thereby mitigating the tendency to align with incorrect instances.

4.3.3 Sensitivity of Weighting Hyperparameters

We conducted experiments with different values of λ_{ssl} ranging from 0.3 to 0.7. As shown in Fig. 5, the results across three different settings in the Office-Home dataset indicate that the self-supervised loss is not sensitive to this hyperparameter.

5 Related Work

Universal Domain Adaptation. UniDA is a more generalized form of DA that makes no assumptions about the label sets relationship between the source and target domains. Prior works [31, 9, 16, 28, 6] utilized score functions (e.g. confidence score, domain transferability) of the target examples from the trained classifier to obtain uncertainty measurements. These uncertainty measurements are crucial for tasks like partial alignment and private class discovery. In these contexts, examples that have high uncertainty are classified as private classes and are consequently assigned lower weights during the alignment process. In the private class discovery task, however, these methods use a manually-designed threshold to detect private classes, which is not applicable in real world scenario. To address this, several methods have been introduced: OVANet [27] and MLNet [19] utilized a binary one-versus-all classifier that distinguishes between common and private classes. Meanwhile, UniOT [4] proposes using the statistical mean to dynamically calculate these

Method	H-score	DoEM
$\mathcal{L}_s + \mathcal{L}_{\text{ssl}}^{500} + \mathcal{L}_{\text{adv}}$	51.5	2.6
$\mathcal{L}_s + \mathcal{L}_{\text{ssl}}^{1000} + \mathcal{L}_{\text{adv}}$	52.3	1.7

Table 5: Studying SSL for the alignment process. $\mathcal{L}_{\text{ssl}}^{\bullet}$ means number of 500 or 1000 training steps for SSL loss (5). At default, we train 1000 steps for the classification loss $\mathcal{L}_s$ (2) and the adversarial loss $\mathcal{L}_{\text{adv}}$ (4) is added at the 500th step. **DoEM** means the degree of explicit misalignment.

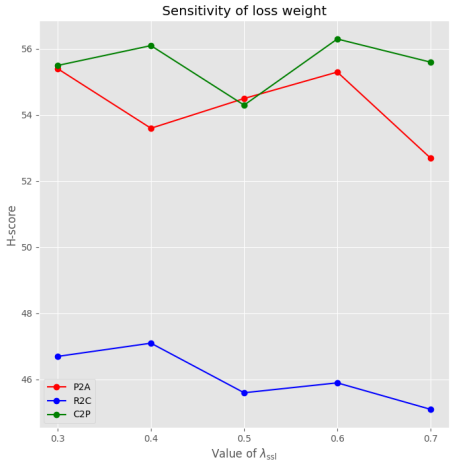


Figure 5: Sensitivity to the weighting hyperparameter λ_{SSL}.

thresholds, thereby adding a level of automation to the process. Similarly, MATHS [5] utilizes Hartigan's dip test to robustly determine the optimal threshold by employing the bimodality properties of the uncertainty scores.

Self-supervised Learning in Domain Adaptation. Self-supervised Learning (SSL) has been used in domain adaptation literature [30, 1]. However, prior works have not clearly explained the rationale for utilizing SSL in domain adaptation. STOC [11] demonstrated that leveraging self-training (ST) on a feature space pretrained with contrastive learning (CL) can yield superior performance compared to using either method in isolation. Their findings indicate that a feature space developed through CL can more effectively capture invariant features, thereby reducing discrepancies. However, applying CL in the pretraining stage requires substantial data and extra computational costs, which limits its application in small datasets. In contrast to prior works, we explore the misalignment issues from *implicit* and *explict misalignment* aspects. Our research is the first work to show that *implicit misalignment* can be aggravated in Extreme UniDA. We further show that SSL can effectively address this problem.

6 Conclusion

In this paper, we introduce the concept of *Extreme UniDA* settings, which emphasize the coverage of source-private classes. Our findings reveal that Extreme UniDA exacerbates misalignment, as indicated by the proposed cross-domain alignment score (CDAS). To understand how misalignment occurs in UniDA, we dissect the issue into *implicit misalignment* and *explicit misalignment*. We further identify that previous works fail to address Extreme UniDA effectively, as they cannot adequately tackle the problem of implicit misalignment, where features are severely entangled prior to alignment. To overcome this limitation, we propose *incorporating target data before the alignment process*. Specifically, we introduce a source-label-free target incorporation module using self-supervised learning (SSL). Our analysis suggests that this approach can effectively reduce implicit misalignment in the initial stage and further alleviate explicit misalignment during the alignment process.

7 Limitation

Our primary approach involves incorporating target data prior to the alignment process to mitigate implicit misalignment. Given the absence of labels in the target data, we utilized self-supervised learning to integrate the target information. However, we acknowledge that there are many potential strategies for leveraging target data that we have not yet explored, and we leave these for future study.

References

[1] Idan Achituve, Haggai Maron, and Gal Chechik. Self-supervised learning for domain adaptation on point clouds. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pages 123–133, 2021.

[2] Shai Ben-David, John Blitzer, Koby Crammer, and Fernando Pereira. Analysis of representations for domain adaptation. *Advances in neural information processing systems*, 19, 2006.

[3] Zhangjie Cao, Lijia Ma, Mingsheng Long, and Jianmin Wang. Partial adversarial domain adaptation. In *Proceedings of the European conference on computer vision (ECCV)*, pages 135–150, 2018.

[4] Wanxing Chang, Ye Shi, Hoang Duong Tuan, and Jingya Wang. Unified optimal transport framework for universal domain adaptation. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho, editors, *Advances in Neural Information Processing Systems*, 2022. URL `https://openreview.net/forum?id=RTan64GlCLV`.

[5] Liang Chen, Qianjin Du, Yihang Lou, Jianzhong He, Tao Bai, and Minghua Deng. Mutual nearest neighbor contrast and hybrid prototype self-training for universal domain adaptation. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(6):6248–6257, Jun. 2022. doi: 10.1609/aaai.v36i6.20574. URL `https://ojs.aaai.org/index.php/AAAI/article/view/20574`.

[6] Liang Chen, Yihang Lou, Jianzhong He, Tao Bai, and Minghua Deng. Evidential neighborhood contrastive learning for universal domain adaptation. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(6):6258–6267, Jun. 2022. doi: 10.1609/aaai.v36i6.20575. URL `https://ojs.aaai.org/index.php/AAAI/article/view/20575`.

[7] Xinlei Chen and Kaiming He. Exploring simple siamese representation learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 15750–15758, 2021.

[8] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009.

[9] Bo Fu, Zhangjie Cao, Mingsheng Long, and Jianmin Wang. Learning to detect open classes for universal domain adaptation. In *European Conference on Computer Vision*, pages 567–583. Springer, 2020.

[10] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario March, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of machine learning research*, 17(59):1–35, 2016.

[11] Saurabh Garg, Amrith Setlur, Zachary Lipton, Sivaraman Balakrishnan, Virginia Smith, and Aditi Raghunathan. Complementary benefits of contrastive learning and self-training under distribution shift. *Advances in Neural Information Processing Systems*, 36, 2024.

[12] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016.

[13] Xiang Jiang, Qicheng Lao, Stan Matwin, and Mohammad Havaei. Implicit class-conditioned domain alignment for unsupervised domain adaptation. In *International conference on machine learning*, pages 4816–4827. PMLR, 2020.

[14] Daniel Kifer, Shai Ben-David, and Johannes Gehrke. Detecting change in data streams. In *VLDB*, volume 4, pages 180–191. Toronto, Canada, 2004.

[15] Jogendra Nath Kundu, Suvaansh Bhambri, Akshay R Kulkarni, Hiran Sarkar, Varun Jampani, et al. Subsidiary prototype alignment for universal domain adaptation. *Advances in Neural Information Processing Systems*, 35:29649–29662, 2022.

[16] Omri Lifshitz and Lior Wolf. Sample selection for universal domain adaptation. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(10):8592–8600, May 2021. doi: 10.1609/aaai.v35i10.17042. URL `https://ojs.aaai.org/index.php/AAAI/article/view/17042`.

[17] Hong Liu, Jianmin Wang, and Mingsheng Long. Cycle self-training for domain adaptation. *Advances in Neural Information Processing Systems*, 34:22968–22981, 2021.

[18] Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I Jordan. Conditional adversarial domain adaptation. *Advances in neural information processing systems*, 31, 2018.

[19] Yanzuo Lu, Meng Shen, Andy J Ma, Xiaohua Xie, and Jian-Huang Lai. Mlnet: Mutual learning network with neighborhood invariance for universal domain adaptation. In *AAAI*, 2024.

[20] Ke Mei, Chuang Zhu, Jiaqi Zou, and Shanghang Zhang. Instance adaptive self-training for unsupervised domain adaptation. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXVI 16*, pages 415–430. Springer, 2020.

[21] Sinno Jialin Pan and Qiang Yang. A survey on transfer learning. *IEEE Transactions on knowledge and data engineering*, 22(10):1345–1359, 2009.

[22] Pau Panareda Busto and Juergen Gall. Open set domain adaptation. In *Proceedings of the IEEE international conference on computer vision*, pages 754–763, 2017.

[23] Xingchao Peng, Ben Usman, Neela Kaushik, Judy Hoffman, Dequan Wang, and Kate Saenko. Visda: The visual domain adaptation challenge. *arXiv preprint arXiv:1710.06924*, 2017.

[24] Xingchao Peng, Qinxun Bai, Xide Xia, Zijun Huang, Kate Saenko, and Bo Wang. Moment matching for multi-source domain adaptation. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 1406–1415, 2019.

[25] Peter J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987. ISSN 0377-0427. doi: https://doi.org/10.1016/0377-0427(87)90125-7. URL `https://www.sciencedirect.com/science/article/pii/0377042787901257`.

[26] Kate Saenko, Brian Kulis, Mario Fritz, and Trevor Darrell. Adapting visual category models to new domains. In *European conference on computer vision*, pages 213–226. Springer, 2010.

[27] Kuniaki Saito and Kate Saenko. Ovanet: One-vs-all network for universal domain adaptation. In *Proceedings of the ieee/cvf international conference on computer vision*, pages 9000–9009, 2021.

[28] Kuniaki Saito, Donghyun Kim, Stan Sclaroff, and Kate Saenko. Universal domain adaptation through self supervision. *Advances in neural information processing systems*, 33:16282–16292, 2020.

[29] Hemanth Venkateswara, Jose Eusebio, Shayok Chakraborty, and Sethuraman Panchanathan. Deep hashing network for unsupervised domain adaptation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 5018–5027, 2017.

[30] Jiaolong Xu, Liang Xiao, and Antonio M López. Self-supervised domain adaptation for computer vision tasks. *IEEE Access*, 7:156694–156706, 2019.

[31] Kaichao You, Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I. Jordan. Universal domain adaptation. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2019.

	General	Extreme
Office-31 [26]	10: 10: 11	24: 5: 3
Office-Home [29]	5: 10: 50	50: 10: 5
Visda [23]	3: 6: 3	8: 2: 2
DomainNet [24]	50: 150: 145	250: 50: 45

Table 6: Comparison of class ratios in the general setting and the extreme setting. It is shown as $|\overline{\mathcal{C}}_s| : |\mathcal{C}| : |\overline{\mathcal{C}}_t|$.

A Supplementary for Experimental Settings

This section provides detailed settings in our implementations and experiments.

A.1 Evaluation Metric

CMU [9] proposed the H-score to highlight the importance of accurately classifying shared instances and detecting private instances. It is calculated as the harmonic mean of the accuracy on common classes and the accuracy on target-private (unknown) classes:

$$\text{H-score} = 2 \cdot \frac{a_c \cdot a_{\overline{c}^t}}{a_c + a_{\overline{c}^t}}$$

where a_c and $a_{\overline{c}^t}$ are the accuracy on common classes and target-private (unknown) classes, respectively.

A.2 Extreme UniDA setting

We set a significant proportion of source-private classes to over 70% to simulate Extreme UniDA. Table 6 shows the ratio of source-private class set $\overline{\mathcal{C}}_s$, common class set $\mathcal{C}$, and target-private class set $\overline{\mathcal{C}}_t$ for general UniDA and Extreme UniDA problems.

A.3 A-distance calculation

$\mathcal{A}$-distance [14] is a classic way to measure the discrepancy between domains:

$$d_A(P, P') = 2 \sup_{A \in \mathcal{A}} |P(A) - P(A')|, \tag{7}$$

where $\mathcal{A}$ is a collection of measurable sets, and P, P' are probability distribution overs the measure space. While it can not be directly calculated, it has been shown to be approximated with a linear classifier h:

$$d_A(\mathcal{D}_s, \mathcal{D}_t) = 2(1 - 2 \min_{h \in H} \text{err}(h)). \tag{8}$$

where the classifier h achieves the minimum error in the binary classification problem of discriminating features from different domains.

A.4 Aggregation score calculation

We utilize K-means clustering combined with silhouette analysis [25] to assess the degree of aggregation within same-class clusters. Initially, we apply K-means clustering to the features within each class. Subsequently, we calculate the silhouette score for these clusters. A higher silhouette score indicates that the data points are well-matched to their own clusters and poorly matched to neighboring clusters, suggesting a high degree of local aggregation.

B Details of Partial Alignment Module

We utilize the feature distance relationships to calculate the weight for each example in feature alignment. Specifically, we compute the cross-domain distance from each example in one domain to

	Office(10/10/10)						
	A2D	A2W	D2A	D2W	W2A	W2D	Avg
UAN [31]	59.7	58.6	60.1	70.6	60.3	71.4	63.5
CMU [9]	68.1	67.3	71.4	79.3	72.2	80.4	73.1
DANCE [28]	72.6	62.4	63.3	76.3	5 7.4	82.8	66.6
UniOT [4]	**87**	**88.5**	**88.4**	**98.8**	**87.6**	96.6	**91.1**
Ours	85.8	83.5	84.7	96.4	84.2	**97.2**	88.6

Table 7: H-score(%) on **Office** (10/10/11)

	Office-Home (5/10/50)												
	Ar2Cl	Ar2Pr	Ar2Rw	Cl2Ar	Cl2Pr	Cl2Rw	Pr2Ar	Pr2Cl	Pr2Rw	Rw2Ar	Rw2Cl	Rw2Pr	Avg
UAN [31]	51.6	51.7	54.3	61.7	57.6	61.9	50.4	47.6	61.5	62.9	52.6	65.2	56.6
CMU [9]	56	56.9	59.2	67.0	64.3	67.8	54.7	51.1	66.4	68.2	57.9	69.7	61.6
DANCE [28]	26.7	11.3	18	33.2	12.5	14.3	41.6	39.9	33.3	16.3	27.1	25.9	25
UniOT [4]	**67.3**	**80.5**	**86.0**	**73.5**	**77.3**	**84.3**	**75.5**	**63.3**	**86.0**	**77.8**	**65.4**	**81.9**	**76.6**
Ours	53.8	75.1	83.9	63.2	67	77.6	72.2	55.9	81.6	74.2	55.9	81.6	70.2

Table 8: H-score(%) on **Office-Home** (5/10/50) .

the nearest neighbors of the alternate domain. We aggregate these distances for all examples, resulting in a bimodal distribution. This occurs because examples from common classes tend to be closer than those from private classes. We then assign lower weights for examples with larger distances.

For private-class discovery, we follow prior work [5] to automatically find the threshold with a Gaussian Mixture Model.

C More Experimental Results

In this section, we reveal the performance of the general UniDA settings (as mentioned in Table 6) and repetitive trials on Office31.

C.1 Results on General UniDA

We report the results of our method in general settings using the Office and Office-Home datasets in Table 7 and Table 8. Our approach outperforms existing methods that utilize the adversarial training paradigm. Although UniOT, which employs the optimal transport paradigm, generally surpasses our method, it struggles in extreme cases, as detailed in the main content. Overall, our method demonstrates better robust performance across various ratios.

C.2 Results with Error Bar

We report the results on Office31 based on three runs in Table 9, each using a different random seed. The standard deviation values are relatively minor compared to the advantages we observe over prior works. The results indicate that our method is stable in repetitive trials.

	A2D	A2W	D2A	D2W	W2A	W2D
H-score	75.6 ± 0.7	85.4 ± 2.6	87.1 ± 2.6	76.3 ± 1.4	72.9 ± 1.3	80.0 ± 2.5

Table 9: H-score of our proposed method with error bars on Office31.